

Track Number: 1

India's Growth Path in BRICS: Evidence from Convergence Analysis from Key Economic Indicators

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Undertaking

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Abstract

With BRICS, a bloc comprising India, Russia, China, South Africa, and Brazil, aiming to position itself as an alternative economic institution and foster economic growth and cooperation, especially in the Global South, analysing its internal economic cohesion is critical to understand its long-term stability. Analysing India's economic position and trajectory is also necessary as it is not only one of the fastest growing economies but also aims to position itself as a military and political powerhouse in the world. The current paper utilises median comparison, sigma convergence, beta convergence, and club convergence tests for these purposes. The main variables used are per capita GDP, GDP growth rate, trade openness, and FDI Inflow.

The empirical results provide robust evidence of bloc-wide sigma convergence of all the variables except FDI Inflows, representing a 27.65% reduction in inequality. However, in terms of per capita GDP, the analysis found two clubs. A high-income club comprising China and Russia, and an emerging economy club comprising South Africa, Brazil, and India. The paper identifies India as the strongest convergence driver due to consistently recording the highest growth rates while being the poorest country in the bloc. It also narrowed its per capita GDP to 50% of the median and was successful in opening its economy and attracting foreign investment during global shocks, thereby reducing intra-bloc inequality.

Introduction

Since its formal inception in 2006, BRICS — a bloc grouping Brazil, Russia, India, China, and South Africa — has emerged as one of the most important partnerships of emerging economies, collectively accounting for a significant share of global output, trade, and investment. Often compared and analysed as a counterbalance to established groups, like G6 and EU, and advanced economies, significant differences persist across members in terms of income levels, growth models, and structural drivers of development (O'Neill, 2001). Understanding whether BRICS countries exhibit economic convergence is therefore central to assessing the bloc's internal cohesion and long-run sustainability, which this study aims to do (Barro & Sala-i-Martin, 1992). It becomes even more necessary to understand internal economic alignment when the group seeks to institutionalize its economic cooperation, by funding developmental projects and providing liquidity support, through mechanisms such as the New Development Bank (NDB) and the Contingent Reserve Arrangement (CRA).

Economic convergence studies generally consist of three tests: σ -convergence, which examines whether the dispersion of income or indicators declines over time; β -convergence, which tests whether poorer economies grow faster than richer ones; and club convergence, which allows for the possibility of multiple steady states where only subsets of countries converge. While these frameworks have been extensively applied to regional groups such as the European Union and OECD, existing studies on BRICS often view the bloc in aggregate, overlooking member-specific dynamics. This study departs from traditional aggregate analysis to evaluate if the bloc possesses the internal economic cohesion required to function as a unified entity.

Prior studies highlight that convergence results within BRICS are mixed and sensitive to methodology, with some finding only partial or conditional convergence, and many emphasizing dependencies on structural variables and growth drivers (Banday et al., 2021; Lohani, 2021; Das & Nayak, 2023; Phiri, 2018). Yet there remains a notable gap in the literature: few analyses have explicitly focused on India's trajectory within BRICS, evaluating its convergence with the bloc (Kumar et al., 2023). India represents not only one of the fastest-growing large economies but also one of the most populous nations (Export-Import Bank of India, 2016). India's emerging position as a military and political powerhouse adds reason to analysing its relative economic positioning and trajectory. Addressing this gap is important both for scholarly understanding of convergence theory, for practical

assessments of BRICS' internal economic balance and formulating bloc-level policies on economic cooperation, investment flows, and trade through this understanding.

The present study contributes to the existing literature by tackling gaps along multiple dimensions. Using GDP per capita (PPP) as the core income measure, supplemented by growth rate, FDI inflows, and trade openness as complementary indicators, the study uses descriptive benchmarking, σ - and β -convergence tests, and Phillips–Sul club convergence methods. The main aim of this research is to assess India's relative positioning within the bloc and assess the bloc's homogeneity, evaluating if India is converging with the group, over time through these key economic indicators. And in doing so, the study advances both the empirical literature on convergence and the understanding of India's evolving role within BRICS.

Literature Review

There are numerous studies defining the various types of convergence and its different methodologies (Islam, 2003; Iranmanesh, 2025). Convergence can be tested in several ways: β -convergence, which tracks if poorer economies grow faster than richer ones; σ -convergence, which tracks if income dispersion is declining across economies; stochastic convergence, which tests whether income differences are stationary over time; and club convergence, which tests if subsets of economies converge even if the full group does not (Bernard & Durlauf, 1995; Phillips & Sul, 2007; Islam, 2003). While σ -convergence offers a clear snapshot of narrowing inequality, β -convergence captures dynamic growth rates, stochastic convergence considers temporal stability, and club convergence highlights potential heterogeneity within economic blocs (Phillips & Sul, 2007). Each approach captures different dimensions of catch-up dynamics and together provides a more complete and comprehensive picture to analyse growth patterns among fundamentally different economies (Islam, 2003).

With 41% of the global population, 24% of the global GDP, and more than 16% of global commerce, BRICS is a significant organisation that brings together the most prominent emerging economies worldwide (Lal, 2023). Its members display substantial differences in economic size, structure, and policy orientation, which complicates direct comparisons and renders group-level averages potentially misleading (Armijo, L. E., 2007).

Empirical evidence for BRICS convergence is mixed. Studies applying stochastic and nonlinear tests report partial or conditional convergence, often with stronger results for China and Brazil than for India, Russia, or South Africa (Phiri, 2018). Other works suggest that convergence is sensitive to structural variables such as investment, trade openness, and foreign direct investment (Banday et al., 2021; Lohani, 2021), indicating that convergence outcomes cannot be fully understood without considering these growth drivers. Recent research further suggests that convergence may be club-based, with clubs moving together towards different equilibrium levels (Das & Nayak, 2023).

The research critiques the limitations of aggregate-level analyses and evaluates if BRICS progresses as a cohesive economic whole or fractured unit. The current literature highlights the bloc's heterogeneity and reinforces the need for member-specific analyses, particularly for countries such as India whose growth trajectory may differ. Because BRICS members differ substantially in size, structure, and policy regimes, group-level averages can mask divergent member trajectories; thus, an explicit country-level examination is warranted. The current research aims to focus on India's trajectory and convergence with the bloc, since no literature exists to tackle that gap as far as the researcher's knowledge is concerned. India's growing importance as a military, economic, and political power necessitates this study.

GDP per capita (PPP) is widely used as the core measure for convergence, since it enables cross-country comparability (You et al., 2017; Das & Nayak, 2023; Phiri, 2008). Phiri (2008) correctly identifies the reasoning behind taking our main variable as per capita GDP, stating that whilst BRICS countries collectively boast an impressive combination of market productivity growth levels on a global front, the same cannot be mentioned for their per capita GDP growth rates, which can be described as being moderate at best. The relatively low levels of real GDP per capita associated with BRICS countries are concerning statistics since per capita GDP has been found to be the most adequate measure of economic welfare (Phiri, 2008).

Thus, while GDP per capita is the primary convergence variable, the other indicators namely, trade openness, FDI inflows, and GDP growth rate, supply contextual evidence that helps offer a more nuanced insight into member contribution (Banday et al., 2021; Lohani, 2021). Banday et al. (2021) note that openness in trade minimises the hindrances and allows the import of capital, easy accessibility of technology and development in human capital which fosters economic growth. It is therefore accepted that trade openness and FDI are to be the

key drivers of economic growth in the globalised world. Inclusion of these indicators enables the study to examine not only whether convergence occurs, but also through which channels, providing a more comprehensive understanding of India's positioning relative to its BRICS peers.

Methodology

The data in the present study consists of per capita GDP, GDP growth rate, FDI inflows as a percentage of GDP and Imports and Exports as a percentage of GDP (Trade Openness) collected for the five core BRICS members namely, Russia, China, India, Brazil, and South Africa. The data has been collected in annual intervals to cover the period between 2006–2024, capturing the full lifecycle of the bloc to date from the World Bank Group's World Development Indicators database.

The current paper analyses India's position and trajectory in the bloc using median comparison. Median is used instead of mean as it helps to reduce the influence of outliers. Next, convergence was tested in several ways. First, β -convergence was examined to determine whether poorer economies within BRICS grow faster than the richer economies. A panel regression is estimated where the dependent variable is the growth rate of each indicator, and the independent variable is its initial level relative to the BRICS group, testing conditional or absolute catch-up dynamics. Second, σ -convergence was assessed to check whether the cross-sectional dispersion of economic indicators across BRICS members declines over time. Following Barro and Sala-i-Martin (1992), dispersion is measured as the standard deviation of the natural logarithm of each indicator for every year in the sample. Finally, club convergence was tested by employing Phillips and Sul's (2007) log-t regression test to account for heterogeneity and identify subgroups that converge together (Bernard & Durlauf, 1995; Phillips & Sul, 2007; Islam, 2003). As explained earlier, each approach captures different dimensions of catch-up dynamics and together provides a more complete and comprehensive picture to analyse growth patterns among fundamentally different economies (Islam, 2003).

Results and Discussion

As seen in Table 1 and Table 2, we find that India ranked number 1 in GDP growth among all five nations in 2015, 2016, 2021, 2022, 2023, and 2024. It has also been in the top three in GDP growth in the rest of the years with the exceptions of 2008 and 2020. India's per capita GDP ratio with the median increased from 30.6% in 2006 to 50% in 2024. However, India has always ranked last in GDP per capita, making it the poorest country in the bloc. When BRICS was founded, India was a relatively closed economy with its trade openness as 85% of the median but India was able to increase it to 113% of the median by 2024 to position itself as a trading hub. This is corroborated with the findings of Kumar et al. (2023) who found that India's export position has improved relative to its BRICS peers. India also consistently stayed above the median FDI inflows, with the exception of 2010 - 2013, with the latest being 119% of the median in 2024. This shows that India attracted investment even in unfavourable global situations where emerging economies faced FDI outflows.

Sigma convergence showed a strong trend towards per capita GDP homogenization within the bloc as shown in Figure 1. Dispersion decreased from 0.763 in 2006 to 0.552 in 2024, representing a 27.6% reduction in per capita GDP inequality. The downward negative slope confirms that the nations are becoming more equal. Similarly, GDP growth rates also showed homogenization within the bloc as represented in Figure 2. However, dispersion of GDP growth rates exhibited a cyclical pattern. Standard deviation spiked during global crises like the 2008 Great Depression and the 2020 COVID pandemic but settled in lower levels during stable periods. This shows that while each of the economies were affected differently in magnitude, they were all exposed to these shocks irrespective of their economic foundation. A similar pattern was seen, in Figure 3, with trade openness. While it was moving towards homogeneity, standard deviation spiked during major global shocks. This again shows the varying effect of global shocks in BRICS economies and the absence of a common BRICS trading model. This is primarily due to the fundamentally different models and policies adopted by the countries. We see in Figure 4 that FDI Inflows is the only variable showing divergence. While the bloc initially converged from 1.69 in 2006 to 0.4 in 2009, the trend reversed dramatically post 2009. Investors preferred individual countries instead of the bloc as a whole, with a major chunk of capital going towards emerging economies, hence widening the dispersion.

The estimate for $\log_initial$ confirms beta convergence in Table 3. -0.025 confirms that for every 1% richer a country gets, it grows 0.025% slower. This makes the catch-up speed equal to 2.5%. However, because of the extremely small number of countries in consideration ($N = 5$), the p-value (0.196), standard error (0.015), and t-statistic (-1.65) show that the coefficient is statistically insignificant due to small sample bias. It suggests that while beta convergence is directionally correct, the estimates of the magnitude are subject to higher uncertainty due to the sample size being extremely low.

In the Conditional Convergence test shown in Table 4, using the Fixed Effects model to exclude country-specific dynamics, the convergence coefficient increased to -0.048, meaning that poorer countries would grow by 4.8% faster after removing structural differences holding them back. This value is highly significant with more than 99% confidence, as confirmed by the p-value (0.0015). This confirms that beta convergence in BRICS is strong and consistent once structural differences are taken into consideration.

Club convergence found that all the countries were indeed converging to a single equilibrium but only for two variables, namely, trade openness and GDP growth rate. For GDP per capita, it identified two different clubs converging to separate equilibriums, as seen in Table 5. It identified Russia and China as Club 1 while Club 2 was made up of India, Brazil, and South Africa. This showed that Club 1 was converging to a higher equilibrium while Club 2 countries were emerging and played catch-up dynamics at a lower equilibrium. For FDI inflows, Brazil and India were part of Club 1 whereas China and South Africa were part of Club 2 as seen in Table 6. While India was among the lower income club, it is still among the most attractive destinations for capital investment.

Conclusion

Through rigorous data processing and testing, the current literature concludes that BRICS countries are economically converging in most sectors, with the gap between the poorest country and the richest country reducing by 27.7% over the 19-year time period. While the small sample bias shown affects the magnitude of the convergence, it is directionally correct. The complementary convergence tests corroborate the argument by confirming the same trend. With the existence of two convergence equilibriums for per capita GDP and FDI Inflows, we can determine that treating the bloc as a whole is erroneous since internal cohesion is also fractured in parts. Convergence with China is also unlikely in the foreseeable

future. India is the strongest convergence driving economy among the core economies, with it being the only member nation capable of migrating to a mature, high-income economy. This positions India as the key to prevent the bloc from fragmenting into Rich and Poor camps. This also shows a need for a bloc-wide policy understanding if BRICS wants to position itself as an economic powerhouse and counteract existing established economic institutions.

The current literature also provides an insight into the formation of a BRICS currency and developing the New Development Bank by helping to understand the differences in the economies and providing a basis to set country-specific currency weightages. With the current research, policy makers also gain insights into economic bottlenecks and potential development frameworks. India must leverage its FDI inflows to boost manufacturing exports. It must also encourage and aid in the development of MSMEs to align its trade model closer to the developed economies.

Considering different economic structures and its impact on BRICS, the role of BRICS+ economies in strengthening the bloc serve as viable gaps and areas for further research. This will help to formulate targeted policies and more accurately identify economic bottlenecks and solutions.

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Annexure

country	year	rank_gdp_pc	rank_growth	rank_FDI	rank_trade
India	2006	5	3	3	4
India	2007	5	3	4	4
India	2008	5	5	3	4
India	2009	5	2	2	3
India	2010	5	2	4	4
India	2011	5	2	4	1
India	2012	5	2	4	1
India	2013	5	2	5	2
India	2014	5	2	3	2
India	2015	5	1	3	3
India	2016	5	1	3	3
India	2017	5	2	3	3
India	2018	5	2	3	3
India	2019	5	2	3	3
India	2020	5	4	2	3
India	2021	5	1	5	3
India	2022	5	1	3	2
India	2023	5	1	3	2
India	2024	5	1	2	2

Table 1: India's Relative Ranking Within the BRICS Bloc by Indicator

country	year	ratio_gdp_pc	gap_growth	ratio_growth	ratio_trade
India	2006	0.306803	0	1	0.850401
India	2007	0.312091	0	1	0.883576
India	2008	0.310886	-2.0075	0.605925	0.999733
India	2009	0.339708	7.987701	-62.4892	1
India	2010	0.356731	0.969359	1.128763	0.987988
India	2011	0.363242	0.941287	1.218902	1.113684
India	2012	0.374475	1.432301	1.355932	1.175094
India	2013	0.389973	3.381284	2.125286	1.163263
India	2014	0.414688	5.996401	5.241257	1.023448
India	2015	0.436713	6.674392	6.049234	1
India	2016	0.440039	7.591753	12.42386	1
India	2017	0.437201	4.969593	3.721887	1
India	2018	0.433225	3.646606	2.298998	1
India	2019	0.440105	1.673361	1.761285	1
India	2020	0.427039	-2.50097	1.763244	1
India	2021	0.44535	3.823101	1.651684	1
India	2022	0.463209	4.592671	2.522418	1.170454

India	2023	0.487518	5.10823	2.251243	1.101839
India	2024	0.49965	2.139849	1.492556	1.13068

Table 2: India's Indicators as a Percentage of the Group Median

term	estimate	std.error	statistic	p.value
(Intercept)	0.263048	0.141318	1.861391	0.15961
log_initial	-0.02503	0.015102	-1.6575	0.195999

Table 3: Cross-Sectional Estimates of Unconditional β -Convergence

term	estimate	std_error	t_value	p_value
log_gdp_pc_lag	-0.04802	0.014696	3.26726	0.001574

Table 4: Estimates of Conditional Convergence

country	club
China	club1
Russian Federation	club1
Brazil	club2
India	club2
South Africa	club2

Table 5: Convergence Clubs: GDP per Capita

country	club
Brazil	club1
India	club1
China	club2
South Africa	club2

Table 6: Convergence Clubs: FDI Inflows

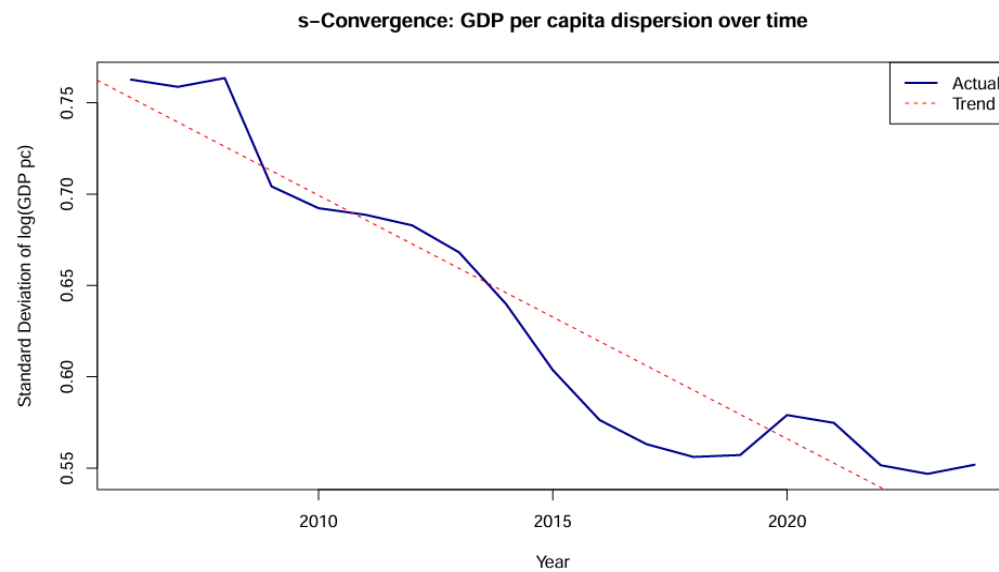


Figure 1: σ -Convergence: GDP per Capita

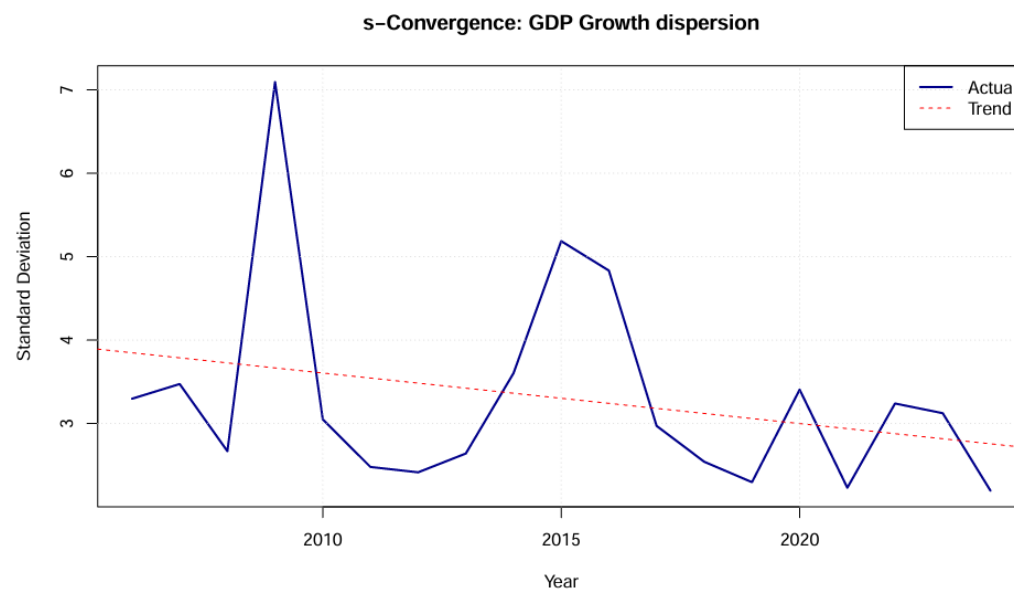


Figure 2: σ -Convergence: GDP growth rate

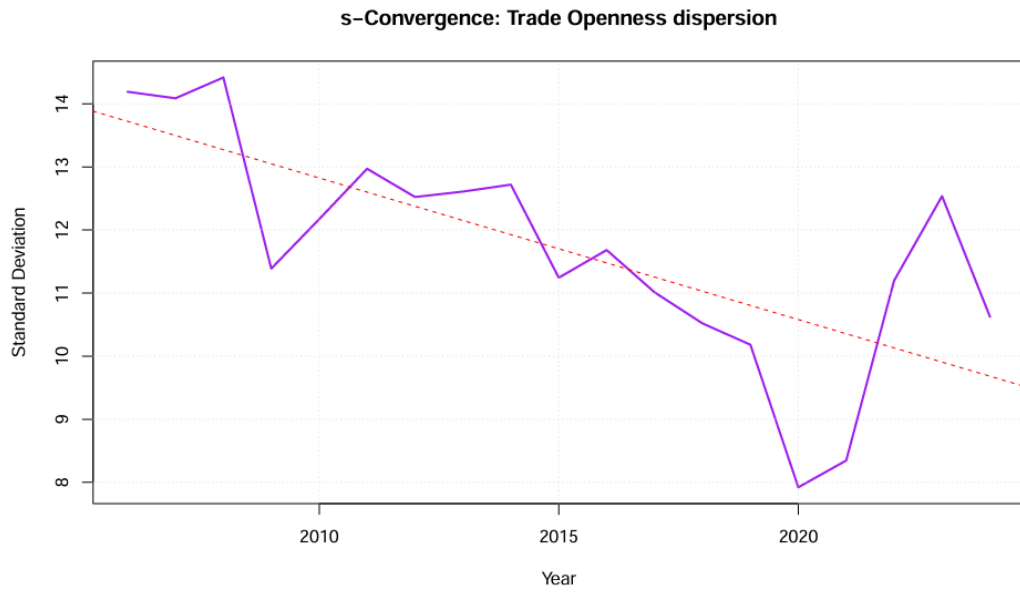


Figure 3: σ -Convergence: Trade Openness

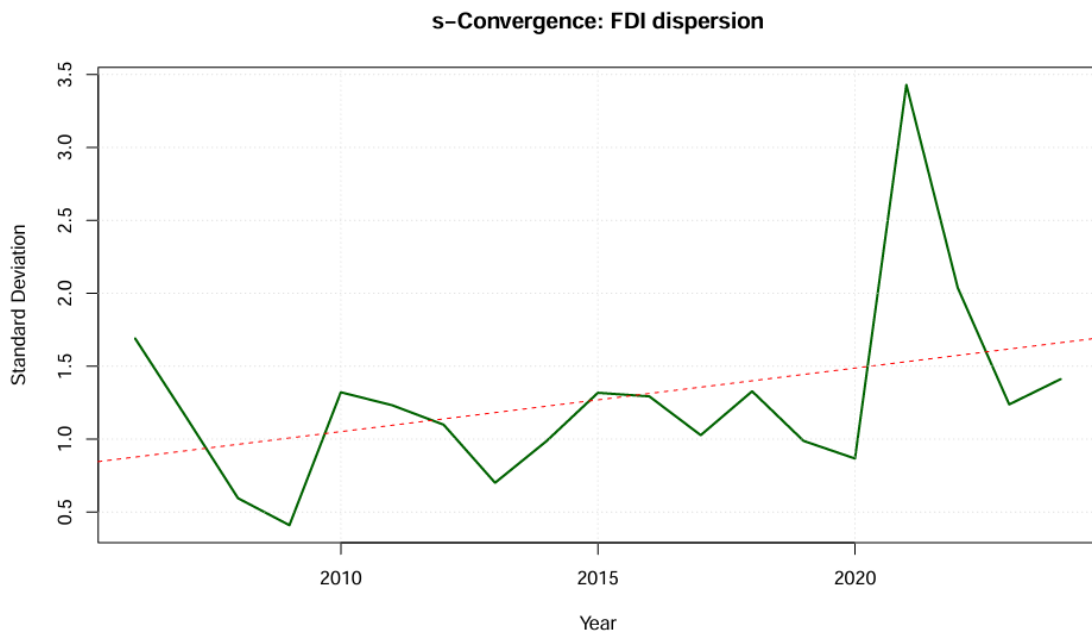


Figure 4: σ -Convergence: FDI Inflows